

ENBC332: Homework #2

Due: Monday, 09/22/25 by noon, beginning of the class

Submit your homework to ELMS, including R code and dataset

Overall points: 50

Note: Late homework may be submitted up to 24 hours after the due date, but a 50% penalty will be applied.

Note: Please aim to write clean, well-organized code that is easy for others to understand. Be sure to include comments or explanations where needed. Points may be deducted if this is not done.

Note: Please name your codes using the following format: "problem1_fullname.R". Start with the problem number and followed by your full name, e.g. "**problem1_jimmyazarnoosh**".

Note: Using generative AI for learning purposes is allowed. However, directly copying or closely replicating AI-generated content is strictly prohibited and will result in a grade of **ZERO**. Repeated violations may lead to more severe consequences.

Note: Please use the discussion board and share with everyone what dataset you are using. Everyone should use unique datasets. Duplicated datasets are not accepted.

Note: Submit a **single zip file** containing the following:

- The dataset you used
- R code files
- Your answers to the questions in PDF format

Problem 1. Find a dataset online (Kaggle) or any other websites. Similar to homework 1, create an Excel file with a dataset that contains only **TWO columns** with numbers. Assign names to the first row for each column. After you decide on your dataset, please share it with others on the discussion board to prevent duplication of dataset used in your HW. So, make sure the dataset you use is unique. Write a R code and calculate the following. For each part make a separate R code. (30 points)

a) Calculate **range** for both columns. (2 points)

b) Calculate **variance** for both columns, using R function "`var()`" and also write your own custom functions. (8 points)

1) Write a function to calculate variance using a **for loop**.

2) Write a function to calculate variance using the **sum()** function.

c) Calculate **standard deviation** for both columns, using R function "`sd()`" and also write your own custom functions. (8 points)

1) Write a function to calculate standard deviation using a **for loop**.

2) Write a function to calculate standard deviation using the **sum()** function.

d) Calculate **quartiles** for both columns, using R function "`quantile()`" and also write your own custom function. (8 points)

e) Calculate **Interquartile (IQR)** for both columns, using R function "`IQR()`" and also write your own custom function. (4 points)

Problem 2. In this exercise, you will practice the multiplication principle. Write R code to calculate the number of ways to fill **r positions** with **n objects**. (10 points)

For **n**, define a random integer number between 5 to 20 ($5 \leq n \leq 20$).

For **r**, define a random integer number between 1 to less than or equal n ($1 \leq r \leq n$).

For example, if you pick 5 apples with 3 positions, your code should calculate the number of ways to fill 3 positions. For reference, see slide 10 of Lecture 6 on probability.

Hint: Use `sample()` to get random integer numbers.

Problem 3. Write R code to calculate the probability of different outcomes when **rolling two dice**. Create a function that takes two inputs: one input represents the outcome from die 1, and the other input represents the outcome from die 2. For reference, see slide 9 of Lecture 7 on distribution and probability theory. (10 points)

Hint: Use `sample()` to get random integer numbers.

Useful comments:

All your functions can follow this format:

e.g. Problem 1a

```
Data <- trees$Height
my_range_func <- function(Data){
  Range <- max(Data) - min(Data)
  return(Range)
}
```

Problem 1b) For each column you calculate variance 3 times. All 3 values should be similar.

Problem 1c) Again, for each column you calculate standard deviation 3 times. You can use your code in 1b and only add the square root of variance to calculate standard deviation.

Problem 1d) `quantile` function in R gives you 5 numbers. Your function should produce same numbers. Numbers represent 0%, 25%, 50%, 75%, and 100% of the data after you sort it. Follow slide 23 in lecture “05_Week2_”.

Problem 1e) Reuse the function you wrote in 1d for quantile and get Q1 and Q3 to calculate IQR.

Problem 2) This is an easy problem just be sure that n is greater than r. You randomly get integer values for n and r ($n > r$).

Problem 3) First define a random number for each die i.e. die1 and die2. Your function may have only 2 inputs, die1 and die2. Define a vector with probabilities corresponding to each outcome listed in the table on slide 9 in lecture “07_Week3_”. I recommend you assign names to each probability using `names()`. For example, assign 2 to probability 0.03. Or assign 7 to probability 0.17. Feel free to use any other approach you like. Ask me questions if you need help with this problem.